

# Hannah Nguyen

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## EDUCATION

### University of California, Santa Barbara

Santa Barbara, California

*Bachelor of Science in Statistics and Data Science and Applied Mathematics*      *Expected Graduation Date: June 2027*

- Relevant Coursework: Probability and Statistics, Design and Analysis of Experiments, Regression Analysis, Theoretical Linear Algebra, Mathematical Methods of Analysis, Partial Differential Equations, Intermediate Python, Intermediate C++, SAS Base Programming
- Statistics and Data Science GPA: 3.78, Applied Math GPA: 3.82, Overall GPA: 3.81

## EXPERIENCE

### Data Scientist Intern

October 2024 – August 2025

*IDXExchange LLC*

*Remote*

- Collaborated on a prediction model as part of a team, then independently took over and continued development after team members exited early.
- Self-taught data cleaning, model building, and evaluation, while balancing coursework in statistics and data science.
- Analyzed large datasets and used problem-solving skills to improve efficiency and accuracy in project outcomes.
- Created a comprehensive guide to assist future interns, documented key methodologies and challenges.

### Note Taker

January 2025 – June 2025

*UCSB Disabled Students Program*

*Santa Barbara, California*

- Provided clear and detailed notes on time, ensuring concepts and formulas were legible.
- Supported multiple students, up to five, across different subjects, including Partial Differential Equations, Probability and Statistics, and Regression Analysis.
- Maintained consistency and accuracy, improved understanding of advanced mathematical and statistical concepts.

## PROJECTS

### Close Price Prediction Model | *Python, Numpy, Pandas, Git*

October 2024 - August 2025

- Developed a property closing price prediction model, using RandomForest, with a focus on hyperparameter tuning and model optimization to improve performance.
- Processed and cleaned numerical data (including outlier removal of top/bottom 1%) to improve data quality and model performance.
- Evaluated with multiple training/testing strategies, best model achieving MAE = 101k, RMSE = 243k, MAPE = 18.21%,  $R^2 = 0.91$ , on filtered test dataset.
- Documented progress and challenges in a detailed guide to assist future interns and ensure project continuity.

### Impact of Paperclip Positions on Final Distance | *R, ggplot2, RStudio*

May 2025 – June 2025

- Conducted a Factorial Design experiment testing the effects of paperclip positions on final distance.
- Performed sample size calculations and used linear modeling in R to conduct statistical analysis on data.
- Visualized and analyzed data using ggplot2 and kable tables.
- Practiced technical writing and clear communication, applying topic to broader context and referencing multiple peer reviewed articles, as well as interpreting results from analyses.

### Regression Analysis on Diamond Prices | *R, ggplot2, RStudio*

May 2025 – June 2025

- Built and interpreted simple and multiple linear regression models, including applying linear transformations.
- Explored relationships between variables through descriptive statistics and visualizations.
- Detected multicollinearity between variables using variance inflation factor (VIF) and correlation matrices.
- Constructed and interpreted confidence and prediction intervals.

## TECHNICAL SKILLS

**Languages:** Python (Intermediate), C++ (Intermediate), R (Intermediate), SAS Base Programming (Intermediate), LaTeX, HTML/CSS (Beginner)

**Tools & Libraries:** Python (pandas, NumPy, scikit-learn, PyTorch, Flask), R (ggplot2, dplyr), SQL, Excel

**IDE/Text editor:** Visual Studio Code, RStudio, Overleaf (for LaTeX)

**Data Science:** Probability and Statistics, Design of Experiments, Regression Analysis, Object-Oriented Programming